

2.3: Modeling with Linear Functions

Example. Elan is a college student who plans to spend a summer in Seattle. Elan has saved \$3,500 for the trip and anticipates spending \$400 each week on rent, food, and activities. Write a linear model to represent this scenario. Identify and interpret:

- the rate of change $\rightarrow -\$400/\text{week}$
- the initial amount $\rightarrow M(0) = \$3500$
- the x -intercept $\rightarrow \text{solve } M(t) = 0$

Let t represent the number of weeks

Let $M(t)$ represent the amount of money remaining

$$\rightarrow M(t) = 3500 - 400t$$

$$0 = 3500 - 400t$$

$$400t = 3500$$

$$t = \frac{35}{4} = 8.75$$

Elan will run out of money after 8.75 weeks

Domain! $0 \leq t \leq 8.75$

We can assume that $M(t) \leq 0$ does not make sense in this problem

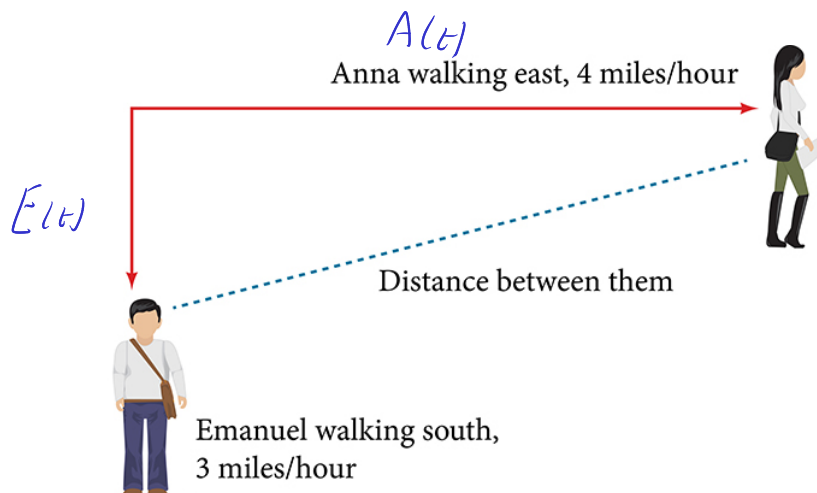
Example. Anne and Emanuel start at the same intersection. Anna walks east at 4 miles per hour while Emanuel walks south at 3 miles per hour. They are communicating with a two-way radio that has a range of 2 miles. How long after they start walking will they fall out of radio contact?

$$[D(t)]^2 = [E(t)]^2 + [A(t)]^2$$

$$= (3t)^2 + (4t)^2$$

$$= 9t^2 + 16t^2$$

$$= 25t^2$$



$$\Rightarrow D(t) = \pm \sqrt{25t^2} = \pm 5|t| = 5t$$

Because we took a square root

Because t could be negative

Because we only consider nonnegative distances and nonnegative t

Solve $D(t) = 2$

$$5t = 2$$

$$t = \frac{2}{5} \text{ hours} = 24 \text{ minutes}$$

Example. Jamal is choosing between two truck-rental companies. The first, *Keep on Trucking, Inc.*, charges an up-front fee of \$20, then 59 cents a mile. The second, *Move It Your Way*, charges an up-front fee of \$16, then 63 cents a mile. When will *Keep on Trucking, Inc.* be the better choice for Jamal?

Let x represent the miles driven

Let $K(x)$ represent the cost for Keep on Trucking, Inc

Let $M(x)$ represent the cost for Move It Your Way

$$K(x) = 20 + 0.59x$$

$$M(x) = 16 + 0.63x$$

$$\text{Solve } K(x) = M(x)$$

$$20 + 0.59x = 16 + 0.63x$$

$$\frac{4}{0.04} = \frac{0.04x}{0.04}$$

$$100 = x$$

When $x = 100$ miles, $K(x) = M(x)$.

Since $M(x)$ has a steeper slope, we know $K(x) \leq M(x)$ when $x \geq 100$.